L9 : Hermite Interpolation

* কম্ম data দিয়ে more accurate

x	f(x)	f'(x)
x_0		
x_1		
x_2		

Degree, $n = 2$

$(n+1)$ Condition / Node / Basis

This basis helps to hold more information

Important

$$P_{2n+1}(x) = h_0(x)f(x_0) + h_1(x)f(x_1) + h_2(x)f(x_2) + \hat{h}_0 f'(x_0) + \hat{h}_1 f'(x_1) + \dots + \hat{h}_n f'(x_n)$$

$$h_k(x) = \{1 - 2(x - x_k) l_k'(x_k)\} l_k(x)^2$$

$$\hat{h}_k = (x - x_k) l_k(x)^2$$

$l_k'(x_k)$

① $l_k(x_0)$

② $l_k'(x)$

③ $l_k'(x_k)$ Last এ

$f(x) = \sin x$, $x \in [0, \frac{\pi}{2}]$; Make table.

Ans:

x	f(x)	$f'(x) = \cos x$
0	0	1
$\frac{\pi}{2}$	1	0

simplify ⇒ $h_k(x)$, $\hat{h}_k$ এর formula P_{2n+1} এ বসাবো।
0 remove করবো না

	x	f(x)	f'(x)
x_0	0	1	0
x_1	1	0	1
x_2	2	2	0

$$\therefore P_{2n+1}(x) = P_5 = h_0(x)f(x_0) + h_1(x)f(x_1) + h_2(x)f(x_2) + \hat{h}_0(x)f'(x_0) + \hat{h}_1(x)f'(x_1) + \hat{h}_2(x)f'(x_2)$$

$$= h_0(x) + 2h_2(x) + \hat{h}_1$$

For $h_0(x)$

$$h_0(x) = \{1 - 2(x-x_0)l_0'(x_0)\}\, l_0(x)^2$$

$$= \left\{1 - 2x \times \left(-\tfrac{3}{2}\right)\right\} \left(\tfrac{1}{2}(x-1)(x-2)\right)^2$$

$$= \frac{1}{4}(3x+1)(x-1)^2(x-2)^2$$

Hermite basis

$$l_0(x) = \frac{x-x_1}{x_0-x_1} \times \frac{x-x_2}{x_0-x_2} = \frac{1}{2}(x-1)(x-2)$$

$$l_0'(x) = \frac{1}{2}(2x-3)$$

$$l_0'(0) = -\frac{3}{2}$$

For $h_2(x)$

$$h_2(x) = \{1 - 2(x-x_2)l_2'(x_2)\}\, l_2(x)^2$$

$$= \left\{1 - 2(x-2) \times \frac{3}{2}\right\} \left(\frac{1}{4}x^2(x-1)^2\right)$$

$$= (-1-3x)\,\frac{1}{2}x^2(x-1)^2$$

$$l_2(x) = \frac{x-x_0}{x_2-x_0} \times \frac{x-x_1}{x_2-x_1} = \frac{1}{2}x(x-1)$$

$$l_2'(x) = \frac{1}{2}(2x-1)$$

$$l_2'(2) = \frac{3}{2}$$

$$\hat{h}_1 = (x - x_1)\, l_1(x)^2$$

$$= (x-1)\left\{\frac{x(x-2)}{-1}\right\}^2$$

Lagrange basis

$$l_1(x) = \frac{x - x_0}{x_1 - x_0} \times \frac{x - x_2}{x_1 - x_2}$$

$$= \frac{x(x-2)}{-1}$$

Exam এ বলে দেয় $h_2(x)$, $\hat{h}_1$ শুধু বের করতে। পুরাটা চায় না normally

[Spring 26] Simplify and then compare with Taylor series / Lagrange

Problem $\xrightarrow{\text{solve with Hermite}}$ Equation $\xrightarrow{\text{Compare with Taylor/Lagrance}}$ Coefficient

For lagrange

$x_0 = 0, 1, 2$